

SQL

MBUA 512

Class 3 – SQL

SQL = Structured Query Language, "es-queue-el" or "sequel". With Michael Winikoff.

Recap from last week (database design):

1. Data modelling using diagrams
2. Translating diagrams to relational databases, and the role of *keys*
3. Anomalies, and avoiding them by *normalisation*

Based on slides by Professor Markus Luczak-Roesch.

Some useful Superset commands

```
show schemas from idrive
show tables in idrive.relbench_rel_ratebeer  -- most tables
show tables in idrive.relbench_rel_avito
select * from idrive.relbench_rel_ratebeer.countries
```

Select schema `user_[username]` , replacing `[username]` with your username
(e.g. for me, `user_michaelwinikoff`).

SQL commands

- `CREATE TABLE` and `ALTER TABLE` - define the structure of a table (its *schema*)
- `INSERT INTO` and `DELETE FROM` and `UPDATE` - change the data in the table
- `SELECT` - query data

(there are a lot more than these)

CREATE TABLE

```
CREATE TABLE name (  
    column_name data_type,  
    column_name data_type,  
    ...  
)
```

To do: how do we deal with keys?

Trino types

- Boolean – `true` or `false`
- Integer / Int – `12` , `-14`
- Real, Double, Number – `3.14` , `-9.8345` (floating point, inexact)
- Decimal – `3.29` (exact, up to 38 digits)
- Varchar – a string, `'Hello!'` (variable-length characters; use single quotes)
- Varbinary – binary data
- Date – `DATE '2026-12-31'` (year-month-day, with the word `DATE`)
- Time – `TIME '01:58:36.77'` (with the word `TIME` ; also a time-with-time-zone)

For MySQL: don't use the `DATE` keyword the same way, and give `varchar` a length. ANSI SQL: `INTEGER` , `REAL` , `DECIMAL` , `CHARACTER` , `CHARACTER VARYING` .

CREATE TABLE – example

```
CREATE TABLE countries (  
    code VARCHAR(50),  
    name VARCHAR(50)  
);
```

trino.io/docs/current/sql/create-table.html

ALTER TABLE

`ALTER TABLE` changes the definition of an existing table.

```
ALTER TABLE name ADD COLUMN column_name data_type
```

```
ALTER TABLE countries ADD COLUMN continent VARCHAR
```

trino.io/docs/current/sql/alter-table.html

INSERT INTO

```
INSERT INTO table_name (columns...) VALUES (vals...), ...
```

Single quotes are for strings, **double quotes** for database objects.

```
INSERT INTO countries (code, name, continent) VALUES
('A7', 'Kepler Dome', 'Moon'),
('A8', 'Blip', 'Moon'),
('A9', 'Blah', 'Moon'),
('A10', 'Blip', 'Moon'),
('A11', 'Blah', 'Moon'),
('A12', 'Blip', 'Moon');
```

To generate rows from a CSV: ask an LLM, or use a spreadsheet with a few formulae (next slide).

Using a spreadsheet

code	name	continent	INSERT INTO countries VALUES			
A7	Kepler Dome	Moon	'A7'	'Kepler Dome'	'Moon'	('A7','Kepler Dome','Moon'),
A8	Blip	Moon	'A8'	'Blip'	'Moon'	('A8','Blip','Moon'),
A9	Blah	Moon	'A9'	'Blah'	'Moon'	('A9','Blah','Moon'),
A10	Blip	Moon	'A10'	'Blip'	'Moon'	('A10','Blip','Moon'),
A11	Blah	Moon	'A11'	'Blah'	'Moon'	('A11','Blah','Moon'),
			=""&A2&""			
			="(" & TEXTJOIN(";",FALSE,D2:F2) & "),"			

DELETE FROM table

SELECT

```
SELECT what
FROM   table(s)
WHERE  conditions
```

```
-- "--" starts a comment to the end of the line
```

```
/* this is also
   a comment */
```

```
SELECT *                                -- "*" means "everything"
FROM   relbench_rel_ratebeer.countries
WHERE  has_brewer = false
```

trino.io/docs/current/sql/select.html

By far the most complex – basics today, more next week.

Today – SQL

- `CREATE TABLE` and `ALTER TABLE` – define the structure of a table (its *schema*)
- `INSERT INTO`, `DELETE FROM`, `UPDATE` – change the data in the table
- `SELECT` – query data

Next: activities in part 2 today – learn by doing. Next week, more SQL, focusing on `SELECT`.

Activities

Activity – ...

~ 10 minutes, in groups